

EasySave v2.0

Technical Specifications

ProSoft — v2.0 — 2026

1. Context

Following a client satisfaction survey, ProSoft's management has decided to develop EasySave v2.0 with significant improvements: a graphical interface, unlimited backup jobs, file encryption, business software detection, and inherited XML/JSON log format from v1.1.

2. Objectives

- Replace the console interface with a WPF graphical application
- Remove the 5-job limit — allow unlimited backup jobs
- Integrate file encryption via CryptoSoft external tool
- Detect and handle business software conflicts
- Inherit JSON/XML log format selection from v1.1

3. Functional Requirements

3.1 Graphical Interface (WPF)

- The application must use WPF (Windows Presentation Foundation)
- MVVM architecture is mandatory
- Main window: display backup job list with Name, Source, Target, Type
- Buttons: Add, Remove, Run, Run All, Settings
- Real-time progression display using Observer pattern (INotifyPropertyChanged)

3.2 Unlimited Backup Jobs

- No maximum limit on the number of backup jobs
- Jobs must be persisted between sessions (config.json)

3.3 File Encryption (CryptoSoft)

- The application must call CryptoSoft.exe to encrypt files
- Only files with user-defined extensions must be encrypted
- Extensions are configurable in the Settings window
- Encryption time must be logged in ms in the daily log file
- Encryption time values: 0 = no encryption, >0 = ms, <0 = error
- Encryption is applied to the target file after copy

3.4 Business Software Detection

- User can define a business software name in Settings
- If the software is running before backup start: backup is blocked
- If detected during backup: current file completes then backup stops
- Stop event must be written to the log file
- The calculator application (calc) can be used as a substitute for testing

3.5 Log Format (inherited from v1.1)

- User can choose JSON or XML format in Settings

- Format applies to both daily log and state files

3.6 Command Line

- EasySave.exe 1 — execute job 1
- EasySave.exe 1-3 — execute jobs 1 to 3
- EasySave.exe 1;3 — execute jobs 1 and 3

4. Technical Requirements

Requirement	Specification
Language	C# / .NET 8.0
UI Framework	WPF (Windows Presentation Foundation)
Architecture	MVVM + Strategy + Singleton + Observer
Log library	EasyLog.dll (separate project)
Encryption	CryptoSoft.exe (external tool)
Log format	JSON or XML with line breaks
File locations	No c:\temp\ paths — use %AppData%
Code language	English (comments and code)

5. Deliverables

Deliverable	Description
EasySave.exe v2.0	WPF graphical application
EasySave.exe v1.1	Console application (simultaneous delivery)
EasyLog.dll v2.0	Updated library
CryptoSoft.exe	Encryption tool
User Manual v2.0	Updated bilingual documentation
Release Note v2.0	Description of changes
UML Diagrams	Updated class, sequence, use case diagrams
GitHub repository	Tagged v2.0 release